

Test Bench(1) of AND gate

Test Bench of AND Gate:

```
module andgate_tb1;
reg a, b;
wire y;
andgate g1(a, b, y);
initial
begin

$monitor("%b %b %b",a,b,y);
a=0; b=0;

#50 a=0; b=1;
#50 a=1; b=0;
#50 a=1; b=1;
end
endmodule
```

Source Code of AND Gate:

```
module andgate (a, b, y);
input a, b;
output y;
assign y = a & b;
endmodule
```

Test Bench(2) of AND gate

Test Bench of AND Gate:

```
module andgate_tb2;
wire t_y;
reg t_a, t_b;
andgate my_gate(t_a,t_b,t_y);

initial
begin
$monitor("%b %b %b",t_a,t_b,t_y);
t_a <= 1'b0; t_b <= 1'b0;
#50 t_a <= 1'b0; t_b <= 1'b1;
#50 t_a <= 1'b1; t_b <= 1'b0;
#50 t_a <= 1'b1; t_b <= 1'b1;
end
endmodule
```

Source Code of AND Gate:

```
module andgate (a, b, y);
input a, b;
output y;
assign y = a & b;
endmodule

$monitor("a=%b b=%b y=%b",t_a,t_b,t_y);
```

Test Bench(3) of AND gate

Test Bench of AND Gate:

```
module andgate_tb2;
wire t_y;
reg t_a, t_b;
andgate my_gate(t_a,t_b,t_y);

initial
begin
$display("a b c");
$display("-----");

$monitor("a=%b b=%b y=%b",t_a,t_b,t_y);
t_a <= 1'b0; t_b <= 1'b0;
#50 t_a <= 1'b0; t_b <= 1'b1;
#50 t_a <= 1'b1; t_b <= 1'b0;
#50 t_a <= 1'b1; t_b <= 1'b1;
end
endmodule
```

Source Code of AND Gate:

```
module andgate (a, b, y);
input a, b;
output y;
assign y = a & b;
endmodule
```

Test Bench(4- Waveform) of AND gate

Test Bench of AND Gate:

```
module andgate_tb2;
wire t_y;
reg t_a, t_b;
andgate my_gate(t_a,t_b,t_y);

initial
begin
$dumpfile("dump.vcd");

$dumpvars;

$monitor("a=%b b=%b y=%b",t_a,t_b,t_y);
t_a <= 1'b0; t_b <= 1'b0;
#50 t_a <= 1'b0; t_b <= 1'b1;
#50 t_a <= 1'b1; t_b <= 1'b0;
#50 t_a <= 1'b1; t_b <= 1'b1;
end
endmodule
```

Source Code of AND Gate:

```
module andgate (a, b, y);
input a, b;
output y;
assign y = a & b;
endmodule
```